Assignment 3
Total Marks:40

Instructions:

- Show all necessary steps clearly.
- Use proper mathematical notation.
- Assume all selections are uniform unless stated otherwise.
- Round numerical answers to four decimal places where necessary.

Question 1

A survey was conducted among students at a university to understand their preferred specialization track. Each student can prefer exactly one track only. The collected data is shown below:

- **Cybersecurity:** 12 female seniors, 9 male seniors, and 2 male juniors.
- **Data Science:** 7 male seniors, 15 female juniors, and 18 male juniors.
- **Cloud Computing:** 17 female seniors, 13 male seniors, 3 male juniors, and 4 female juniors.

Define the following random variables:

$$G = \text{Gender}, \qquad L = \text{Level}, \qquad F = \text{Field of Interest}$$

where

$$G \in \{\text{Female}, \text{Male}\},$$

$$L \in \{\text{Senior}, \text{Junior}\},$$

and

$$F \in \{\text{Cybersecurity}, \text{Data Science}, \text{Cloud Computing}\}.$$

(i) Construct the full joint probability distribution table for G, L, and F. [4]

(ii) Calculate the probability that a uniformly selected student is **not interested in Data Science**, given that the student is a junior. [3]

(iii) Determine whether the following conditional independence statement is true or false:

$$F \perp L \mid G = \text{Female}.$$

Justify your answer by checking all necessary cases. [4]

(iv) Suppose 25 seniors are selected uniformly from this population. Determine the expected number of selected students who do **not** prefer Data Science. [3]

Question 2

Prove that if

$$P(A \mid B, C) = P(B \mid A, C),$$

then

$$P(A \mid C) = P(B \mid C),$$

assuming all required conditional probabilities are well-defined. [4]

Question 3

A patient, Mr. Rahman, visits a clinic after experiencing fever and severe fatigue. The doctor performs a diagnostic test for a viral infection. The test result comes back positive.

The hospital provides the following information:

- Among truly infected patients, 91% test positive.
- Among truly non-infected patients, 12% test positive.
- Among all visiting patients, 18% are actually infected.

Let

$$D = \text{patient is actually infected},$$

and

$$T^+ = \text{test result is positive}.$$

(a) Given that Mr. Rahman's first test result is positive, calculate the probability that he is actually **not infected**. [5]

(b) Mr. Rahman takes two additional tests using the same machine. The second and third test results are both negative. Assuming that test results are conditionally independent given the true disease status, determine whether the more likely scenario is that Mr. Rahman is infected or not infected.

You must consider all three test results:

$$T^+, T^-, T^-.$$

[5]

Question 4

A real-estate analyst wants to predict house prices based on house size. Let the house size be the input variable x, and the selling price be the output variable y.

The prediction model is defined as

$$h_w(x) = w_1 x + w_0.$$

The dataset is given below:

House Size x sq ft	Selling Price y in lakh taka
1000	50
1500	75
2000	100
2500	125

(a) Interpret the meaning of w_1 and w_0 in the context of linear regression. [2]

(b) The squared error loss for one training example is

$$Loss(h_w) = (y - (w_1 x + w_0))^2.$$

The partial derivatives are given as

$$\frac{\partial Loss(h_w)}{\partial w_0} = -2(y - h_w(x)),$$

and

$$\frac{\partial Loss(h_w)}{\partial w_1} = -2x(y - h_w(x)).$$

Suppose the initial estimates are

$$w_1 = 0.03, \qquad w_0 = 0.$$

Using the first data point only, perform two iterations of gradient descent with learning rate

$$\alpha = 0.000001.$$

Show all calculation steps. [6]

(c) Using the updated values of w_0 and w_1, predict the selling price of a 2100 sq ft house. Compare this prediction with the expected price based on the dataset trend. [3]

(d) Explain why the derivative of the loss function with respect to each weight should be close to zero for achieving better model predictions. [3]

(e) Discuss the possible effect on gradient descent if the learning rate keeps increasing over time. [3]

Question 5

You are training a logistic regression model for binary classification. The model has two input features:

$$y' = \sigma(z),$$

where

$$z = w_1 x_1 + w_2 x_2 + b,$$

and

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

You are given:

$$x_1 = 2, \qquad x_2 = 1,$$

$$y = 1,$$

$$w_1 = 0.3, \qquad w_2 = -0.2,$$

$$b = 0,$$

and

$$\alpha = 0.1.$$

(a) Compute the value of z and the predicted output y'. [3]

(b) Using Binary Cross-Entropy loss, suppose

$$\frac{\partial Loss}{\partial w_i} = (y' - y)x_i.$$

Perform one step of gradient descent to update w_1. [4]

(c) Explain the effect of using a decaying learning rate while training a logistic regression model. [3]

Question 6

In this part, consider a feedforward neural network with one input feature.

The network structure is:

- Input layer: 1 feature, $x_1 = 1$
- Hidden Layer 1: 2 neurons
- Hidden Layer 2: 1 neuron
- Output Layer: 1 neuron
- Activation function: sigmoid at all layers

The sigmoid function is

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

The activation outputs from Hidden Layer 1 are

$$a_1^{[1]} = 0.5, \qquad a_2^{[1]} = 0.8.$$

The weights from Hidden Layer 1 to Hidden Layer 2 are

$$w_1^{[2]} = 0.2, \qquad w_2^{[2]} = 0.3.$$

The bias of Hidden Layer 2 is

$$b^{[2]} = 0.$$

(a) Calculate the weighted input $z^{[2]}$ for Hidden Layer 2. [2]

(b) Calculate the activation output $a^{[2]}$ from Hidden Layer 2. [3]

(c) There is only one input feature, but Hidden Layer 1 produces two activation outputs. Explain why this happens. [3]

End of Assignment